

Ian Leung

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EDUCATION

University of Waterloo

Expected 2029

Bachelor of Computer Science & Statistics

GPA: 3.8/4.0

- President's Scholarship of Distinction (**\$5000**), Undergraduate Achievement Award (**\$1500**)
- Relevant Coursework: Object-Oriented Programming (Advanced), Sequential Programming (Advanced), Functional Programming (Advanced), Computer Architecture, Interpreters (Advanced), Linear Algebra, Data Analytics

EXPERIENCE

Roblox

Jun 2026 – Aug 2026

Software Engineer Intern (Network Reliability) | Go, Python

San Mateo, CA

- Automated the network engineering deployment process, pushing to **5000+** production devices in global datacenters
- Built a GitHub Actions CI/CD pipeline, saving **400+ engineering hours** annually by eliminating manual tasks
- Orchestrated Temporal Workflows with LLM analysis & Slack alerting for safe, autonomous deployment of network changes, preventing outages/misconfigurations and identifying issues on **20+** network devices
- Identified and fixed a resource leak in an internal network service, recovering **7%** CPU utilization across devices

WQC

Jan 2026 – Present

Software Engineer | Rust, Python, Docker

Waterloo, ON

- Launched a Grafana observability platform with **100+** metrics, including membership and infrastructure monitoring
- Harnessed Alertmanager to send automatic Discord alerts for critical service failures, maintaining a **>99.5%** uptime
- Engineered portal.waterlooquantclub.com, a full-stack club recruitment & event portal hosted on AWS for **700+** users
- Built and deployed [2v2bluff](https://2v2bluff.com), a real-time multiplayer **Rust/WebSocket** card game supporting four concurrent players

Sunnybrook Research Institute

Jul 2025 – Aug 2025

Software Research Intern | Python

Toronto, ON

- Built an event tracing, playback, and logging tool that records and simulates clinician workflows in **Focused Ultrasound** software, surfacing **5+** defects and edge cases in real usage patterns
- Automated regression testing by generating synthetic JSON event flows and validating results against recorded baselines, cutting testing time by **91%**

PROJECTS

Jevis – Hack The North 2026 Winner (\$2000) | Python, DOM, JEV, OpenAI

- Engineered an Agentic harness for browser use using JEV, decreasing token cost by **250x** compared to GPT-6 Astra
- Sped up queries on walmart from **5 minutes to 44 seconds** using a DOM-based approach and orchestrator models
- Built a caching layer for repeated agent tasks via precomputed instruction paths, achieving a **15%** cache hit rate
- Designed a web interface for browser use agents, turning simple voice commands into instructions and browser actions

Pharmagotchi – Pharmacy Companion App | Kotlin, Android Studio, Figma, OpenRouter

- Built a Tamagotchi-inspired Android companion app to track health & medications and connect users with physicians
- Integrated an **OpenRouter**-powered chat assistant with Retrieval-Augmented Generation (RAG) that parses health conditions and medications into structured JSON vitals and provides context-aware medication guidance
- Connected Fitbit tracking devices through the Google Health API, centralizing health data for seamless monitoring

PeerAssist – Peer Review Platform | Java, Supabase, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for **20+** students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with Apache PDFBox and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

SKILLS

Languages Python, Java, C++, C#, Go, Kotlin, TypeScript, SQL, Bash

Technologies Temporal, gRPC, REST, PyTorch, Pandas, Numpy, OpenCV, PostgreSQL, React, Node.js

Tools Git, Docker, AWS, Github Actions, Grafana, OpenRouter, Supabase, Android Studio